A Bayesian Replication of the Trade-off Between Personal Cognitive Cost and Social Agent Accuracy

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Abstract

You solve most problems by balancing two options: working things out on your own or asking other people for the solution. Solving the problem on your own is typically more cognitively costly than requesting the solution from others. Thus, you may be more tempted to ask other people for the solution. However, if you believe that the solution provided by others is inaccurate, then you may try even harder to solve the problem on your own (Armitage and Redsaw, 2014). Altogether, there exists a trade-off between the cognitive cost from solving a problem on your own and the accuracy of the solution provided by others. In this paper, we examine whether this trade-off can emerge in a Bayesian model. We presented an agent with a ball-and-urn task. They had to estimate the proportion of white balls in an urn that contained black and white balls. To inform its estimate, this agent could draw one ball at a time from its urn (personal sampling) and ask a “social" agent for the proportion of white balls in its urn (social sampling). Critically, the social agent’s urn composition may differ from the personal urn: in some cases, the social agent’s estimate may be inaccurate. Moreover, the Bayesian agent incurred a cognitive cost for each personal sample. The model combined information from personal and social sampling to inform its overall estimate of the proportion of white balls. Furthermore, this model sought to reduce the uncertainty of its overall estimate and the cognitive cost from personal sampling. The Bayesian agent was exposed to same-urn and different-urn conditions and within each of these conditions, we modulated the cognitive cost of personal sampling. We found that the agent takes more personal samples and fewer social samples when the cognitive cost is lower and the social agent is inaccurate. We also observed that the agent takes fewer personal samples and more social samples when the cognitive cost is higher and the social agent is accurate. All in all, we were able to replicate the cognitive cost and social agent accuracy trade-off.

Introduction

Suppose that you had to solve for λ in

$$0 = \det \begin{pmatrix} (a_{11} + ib_{11} - \lambda) & (a_{12} + ib_{12}) & (a_{13} + ib_{13}) & (a_{14} + ib_{14}) \\ (a_{21} + ib_{21}) & (a_{22} + ib_{22} - \lambda) & (a_{23} + ib_{23}) & (a_{24} + ib_{24}) \\ (a_{31} + ib_{31}) & (a_{32} + ib_{32}) & (a_{33} + ib_{33} - \lambda) & (a_{34} + ib_{34}) \\ (a_{41} + ib_{41}) & (a_{42} + ib_{42}) & (a_{43} + ib_{43}) & (a_{44} + ib_{44} - \lambda) \end{pmatrix}.$$

Would you try to solve it on your own or ask another person for the solution (perhaps a mental math genius)? Suppose that you had to solve for x the following problem: $x = 9 + 9$. Does your problem-solving method change? Crucially, each method has its own advantages and disadvantages. For example, solving the problem on your own typically requires more cognitive effort than asking for the solution from another person. Thus, you may be more tempted to ask another person for the solution to reduce cognitive cost. However, if you believe that the other person is inaccurate, then it may be more worthwhile to solve the problem on your own. Therefore, in problem-solving scenarios where individuals can seek help from another person, behavior is determined by a trade-off between cognitive cost and how much you believe that the other person is accurate. Regarding cognitive cost, humans suffer from cognitive capacity constraints, such as limited working memory capacity and attention bottlenecks (Miller, 1956; Tombu et al., 2011). Therefore, humans seek to reach an optimal outcome whilst minimizing their cognitive cost according to said constraints (Escobar and Mitchell, 2025). For example, suppose that you are presented with a very complex problem and you believe that the other person is accurate. You may be more likely to refer to their answer as it is the easiest path towards reaching the correct answer. Now, suppose that the task is very easy and you believe that the other person is inaccurate. It follows that you may be more likely to solve the problem on your own. These behavioural intuitions were replicated in Armitage and Redsaw's two-phase study (2024). For the first phase, participants were presented with a

screen with 60 white circles arranged in three concentric rings. One, five, or ten of these circles would turn black for 0.75 seconds and return to their original color. After a five-second delay, the participant had to identify which circles turned black. Finally, they received feedback about their performance. This process would be repeated across many trials (See Appendix A). Importantly, increasing the number of circles made the memorization task more cognitively difficult. For the second phase, the participants observed a competent agent and an incompetent agent complete a memory task with cards. These agents were presented with 36 cards; they had to turn over one card at a time to observe their underlying color (See Appendix B). By matching cards with their underlying colors, the agents gained points. The participants would observe the competent agent gain more points than the incompetent one and infer that the former's memory is better. Then, the participants completed the circle memory task again; however, for each trial, the competent or the incompetent agent could assist them with recalling target circles (See Appendix C). Armitage and Redsaw found that participants were more likely to seek help when the difficulty of the task increased and vice-versa regardless of the agent's competency. They also found that individuals were more likely to seek help from the competent agent over the incompetent agent in general. Crucially, they found that participants were more likely to seek help from the competent agent when the difficulty of the task increased. In addition, they found that participants were more likely to solve an easy problem on their own when faced with an incompetent agent. However, we should not expect these results to hold across all problem-solving scenarios where a problem solver can ask for help from others. For example, some individuals may be more interested in learning how to solve the problem; thus, they may be willing to incur a higher cognitive cost even when in the presence of an accurate person. In sum, Armitage and Redsaw results only apply to problem solving scenarios where an individual is solely interested in providing the correct answer.

But how does an individual determine whether another person's judgments are

accurate in the first place? To answer this question, we refer to selective social learning, which is the notion that people prefer to learn information from sources that they believe are accurate (Koenig, Clément, and Harris; 2004). To determine whether a source of information is accurate, humans cross-validate information given by a source with their own understanding of the world when possible. If the source's information converges with their current understanding, then they are more likely to deem this source as accurate and subsequently use its information (Aslin & Newport, 2012; Denison & Xu, 2014). Notably, Koenig, Clément and Harris (2004) provided evidence that infants can determine whether a source of information is accurate. Moreover, they prefer to learn from these accurate informants. In their study, three- and four-year-olds were introduced to two informants: one who consistently labeled familiar objects correctly and one who consistently labeled the same objects incorrectly. After observing both informants label known objects, the children correctly identified which informant was accurate. Then, the children were presented with novel objects and observed how the agents labeled them. Koenig, Clément and Harris (2004) found that children who had flagged an informant as inaccurate during the first phase consistently avoided adopting that informant's labels for the new objects. This cross-validation mechanism gives rise to selective social learning for the most part. Altogether, we have strong evidence that humans (1) evaluate the accuracy of other people by comparing the information they provide against their own understanding of the world; (2) humans are more likely to seek help from another person if they believe that the other person is accurate; (3) humans prefer to solve problems with low cognitive cost on their own; and (4) trade between cognitive cost and external source accuracy: (5) if faced with a complex problem and an accurate source, then humans are more likely to ask for help and (6) if faced with an easy problem and an inaccurate source, then humans are more likely to solve the problem on their own.

How can we formalize these properties into a mathematical model that solves problems? First, most problem-solving scenarios involve inferring some latent variable by

using previous knowledge and observed data; Bayesian models can accomplish this process. Crucially, previous research has discovered Bayesian agents can approximate human behaviour in these inference tasks. (Griffiths et al., 2006; Vul et al., 2014). In particular, Bayesian agents try to infer latent variables by integrating prior knowledge with the information provided by observed data (Griffiths et al., 2024). This logic is encapsulated in Bayes' theorem:

$$P(h \mid d) = \frac{P(d \mid h)\, P(h)}{\sum_{h' \in H} P(d \mid h')\, P(h')} \propto P(d \mid h)\, P(h).$$

Suppose that, given observed data d, there exists a finite set of hypotheses $h \in \mathcal{H}$ that can explain d. We want to determine the probability of a hypothesis h given observed data d, denoted as $P(h|d)$. To accomplish our task, we must understand the different types of probabilities that govern $P(h|d)$. Prior probabilities $P(h)$ specify our degree of belief in each hypothesis h before observing data d. After observing data d, we compute the likelihood $P(d|h)$, which tells us how well hypothesis h explains data d. Then, we multiply $P(h)$ with $P(h|d)$ to compute $\lambda P(h|d)$ where $\lambda \in \mathbb{R}$. Finally, we can use $P(h|d)$ our new prior probability $P(h)$ and repeat this entire process by collecting more data d. To illustrate the mechanics of Bayes' rule, suppose that Albert is drinking some water (d) and we consider three hypotheses that would lead to such behaviour d: Albert is thirsty (h_1), Albert is washing his mouth (h_2), and Albert feels cold (h_3). Regarding the likelihood $P(d|h)$, hypotheses h_1 and h_2 explain d better than hypothesis h_3; therefore, the likelihood is higher for h_1 and h_2 relative to h_3. Concerning the prior $P(h)$, on a daily basis, hypothesis h_1 is more likely than hypothesis h_2 and h_3; therefore, the prior is higher for h_1. All in all, it is more likely that John drank water because he is thirsty.

So far, we have discussed how these Bayesian models can make probabilistic inferences about latent variables. But, in the context of our project, how do they decide whether to solve a problem on their own or to ask another person for the solution? Traditionally, cognitive scientists assume that Bayesian agents choose the action that maximizes the expected utility given a belief state S where S summarizes the outcomes

that the Bayesian agent expects to obtain from action a (Griffiths et al., 2024). Formula-wise, the expected utility of action a, denoted $EU(a|S)$, is defined as:

$$EU(a|S) = \sum_{o \in \mathcal{O}_a} P(o \mid a, S)\, U(o)$$

where $\mathcal{O}_a$ is the set of all possible outcomes from action a, $P(o \mid a, S)$ is the probability of observing outcome o given action a and belief state S, and $U(o)$ is the utility the agent obtains from observing outcome o.

Now, we are left with designing an experimental paradigm that can solicit the six aforementioned properties from our Bayesian model. We used a ball-and-urn task as it is simple to model in a Bayesian manner. Essentially, our Bayesian agent a was faced with an urn that had white and black balls, with the true fraction of white balls (p_w) being unknown; it had to estimate the fraction of white balls in the urn. To achieve its objective, the agent could draw one ball at a time from its own urn (personal sample) or ask for help from another agent (social sample). Regarding the former, agent a would incur a cognitive cost for each personal sample. Regarding the latter, agent a could obtain an estimate from another Bayesian agent b who was estimating the fraction of white balls in a different urn. In one condition, agent b's urn composition was very different from agent a's urn composition (80% white and 30% white, respectively), and in the other condition agent b's urn was identical (30% white for both). Within these conditions, we manipulated the cognitive cost associated with personal sampling. This discrepancy in urn composition across agents a and b and the manipulation of cognitive cost should give rise to the six properties we previously discussed: (1) (2) (3) (4) (5) (6). Structure-wise, Bayesian agent a had 5 rounds to correctly guess the fraction of white balls. Within each round, agent a could perform up to 15 personal and social samples. To mark the end of a round, agent a provided a proctor with their estimate. Then, the proctor told agent a how close their estimate was to the true fraction of white balls. Mechanically speaking, if agent a's

estimate was far from the true proportion of white balls, then it would be more uncertain about its estimate and vice-versa.

Agent a cared about reducing the cognitive cost of personal sampling, decreasing the uncertainty of its estimate, and providing an estimate that was close to the true proportion of white balls. Regarding the former, agent a incurred a fixed cognitive cost from personal sampling as it had to compute the fraction of white balls in its own urn. Agent a also wanted to decrease the uncertainty of its estimate, which can be accomplished by doing personal sampling or social sampling. Altogether, these preferences construct the utility function (1) that will be used to inform the correct action that agent a should take under expected utility maximization.

All in all, we are interested in determining whether a Bayesian model that can make decisions can exhibit behavioural properties: $(1), (2), (3), (4), (5), (6)$. In terms of predictions, we hypothesize that agent a should perform less personal samples as the cognitive cost associated with personal sampling increases. Agent a should perform less social samples as the difference between agent b's urn and agent a's urn increases. When agent a thinks that agent b is accurate and is faced with a high cognitive cost, then agent a should perform less personal samples and more social samples. When agent a thinks that agent b is inaccurate and is faced with a low cognitive cost, then agent a should perform more personal samples and less social samples.

Bayesian Model

In our experimental paradigm, agent a is the only decision-maker. It observes an urn containing white and black balls. However, agent a is unaware of the urn's ball composition. Therefore, agent a wants to estimate the true proportion of white balls in this urn (p_w). Let $X \in \{1, 0\}$ denote the color of a randomly drawn ball from the urn where $X = 1$ denotes a white ball and $X = 0$ denotes a black ball with $\Pr(X = 1) = p$ and $\Pr(X = 0) = 1 - p$. The urn's distribution of balls can be approximated by a Bernoulli distribution with $\mathbb{E}[X] = p$ and $\mathrm{Var}(X) = p(1 - p)$. To better inform agent a's estimate,

agent a can acquire new data by drawing one ball from its urn with replacement (personal sampling) or by asking agent b for its estimate of the proportion of white balls in agent b's urn (social sampling).

Regarding personal sampling, let p_{wa} denote the agent a's belief about the proportion of white balls from personal sampling. $p_{wa} \sim \text{Beta}(\alpha, \beta)$ where $\mathbb{E}[p_{wa}(\alpha, \beta)] = \frac{\alpha}{\alpha+\beta}$ and $\text{Var}(p_{wa}(\alpha, \beta)) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. We use the Beta distribution as it is defined on [0,1], which makes it ideal for modeling probabilities and proportions. Before drawing any balls, agent a assumes a prior of $\alpha = 2$ and $\beta = 2$, corresponding to an expected proportion of $\mathbb{E}[p_{wa}] = \frac{2}{4} = 0.5$ and a variance of $\text{Var}(p_{wa}) = \frac{2 \cdot 2}{(4)^2(5)} = \frac{4}{80} = 0.05$. These values of α and β were selected based on another research project that Andrew is working on: he found that these values approximate the average beliefs of young adults when thinking about possible proportions from an urn containing black and white balls. For example, suppose that agent a believes that $p_{wa} = 0.5$ with $\alpha = 2$ and $\beta = 2$ prior to drawing from the urn. If agent a draws a white ball from the urn, then $\alpha = 2 + 1 = 3$, $\beta = 2 + 0 = 2$, and $p_{wa} = \frac{3}{5}$. Conversely, if agent a draws a black ball from the urn, then $\alpha = 2$, $\beta = 1 + 2 = 3$, and $p_{wa} = \frac{2}{5}$.

Regarding social sampling, let p_{ws} denote agent a's belief about the proportion of white balls from social sampling. $p_{ws} \sim \text{Beta}(\alpha_s, \beta_s)$ where $\mathbb{E}[p_{ws}(\alpha_s, \beta_s)] = \frac{\alpha_s}{\alpha_s+\beta_s}$ and $\text{Var}(p_{ws}(\alpha_s, \beta_s)) = \frac{\alpha_s\beta_s}{(\alpha_s+\beta_s)^2(\alpha_s+\beta_s+1)}$. Crucially, before agent a starts the experiment, agent b has already completed 80 personal samples from its own urn. The 80 personal samples make sure that agent b has a head start on agent a and that agent b is very confident about its estimate, such that agent b's estimate does not significantly change over time. Moreover, for each action that agent a takes, agent b performs one personal sample. When agent a receives agent b's estimate, it receives agent b's current estimate of the proportion of white balls in agent b's urn and the number of personal samples that agent b has taken so far. Before agent a observes agent b's estimate, agent a assumes a prior of $\alpha_s = 2$ and $\beta_s = 2$, corresponding to an expected proportion of $\mathbb{E}[p_{ws}] = \frac{2}{4} = 0.5$ and a variance of

$\text{Var}(p_{ws}) = \frac{2 \cdot 2}{(4)^2(5)} = \frac{4}{80} = 0.05$. These values of α_s and β_s were selected based on another research project that Andrew is working on: he found that these values approximate the average beliefs of young adults when thinking about possible proportions that another person may state. For example, suppose that agent a believes that $p_{ws} = 0.5$ with $\alpha_s = 2$ and $\beta_s = 2$ prior to asking for agent b's estimate. If agent b tells agent a that it has drawn 100 balls thus far and that its estimate is 0.3, then agent a will update its estimate as follows: $\alpha_s = 2 + (0.3)(100) = 32$, $\beta_s = 2 + (1 - 0.3)(100) = 72$, and $p_{ws} = \frac{32}{32+72} = \frac{32}{104} = 0.31$. Suppose that, later on, agent a asks agent b for its estimate again and that agent b indicates that it has drawn 120 balls thus far and that its estimate is 0.2, then agent a updates its beliefs as follows: $\alpha_s = 32 + (0.2)(120) = 32 + 24 = 56$, $\beta_s = 72 + (1 - 0.2)(120) = 72 + 96 = 168$, and $p_{ws} = \frac{56}{168+56} = \frac{56}{224} = 0.25$. Notably, our model fails to remove repeated information; however, given that we are trying to model human behaviour and that humans often commit the same error, this quirk is not a limitation (Enke and Zimmermann, 2017).

Finally, agent a combines the information from personal and social sampling into a mixture distribution S. We assume that agent a is skeptical about whether the social sample observed the same urn. As a result, agent a will weigh the agent b's estimate $\mathbb{E}[p_{ws}]$ proportional to how close $\mathbb{E}[p_{ws}]$ is to agent a's $\mathbb{E}[p_{wa}]$ divided by agent a's uncertainty about their estimate, which is $\text{Var}[p_{wa}]$. The weight on the social sample's p_{ws} will be $w_2 = e^{\frac{-|\mathbb{E}[p_{ws}] - \mathbb{E}[p_{wa}]|}{\text{Var}[p_{wa}]}}$ where $w_1 + w_2 = 1$. This mechanism gives rise to the first behavioural property exhibited by humans (1). For example, if agent a realizes that agent b's estimate strongly deviates from its personal sampling estimate, then it will weigh agent b's input less by reducing w_2. Altogether, we have a mixture distribution $S = w_1 p_{wa} + w_2 p_{ws}$ where $\mathbb{E}[S]$ represents the proportion of white balls that agent a believes are in its urn.

In terms of structure, agent a faces multiple rounds with nested steps where it can estimate p_w. Therefore, all variables will have a two-dimensional time component where r denotes the number of rounds and t denotes the number of steps within each round. These

time subscripts permeate across other variables as well. Within each round r, agent a can perform different actions for each time step t. Let $A = \{a_{ps}, a_{ss}, a_{sp}\}$ denote the set of actions that agent a can take where a_{ps} denotes drawing one ball from their personal sample, a_{ss} denotes obtaining agent b's estimate, and a_{sp} denotes agent a stating the proportion of white balls that it believes is in its urn ($\mathbb{E}[S]$). a_{sp} will be discussed later on. Moreover, after selecting an action $a \in A$, agent a observes an outcome. $\mathcal{O}_{ps}, \mathcal{O}_{ss}, \mathcal{O}_{sp}$ correspond to possible outcomes from personal sampling, social sampling, and stating S, respectively. By construction, there exists $\mathcal{O}_a$ for each $a \in A$, such that $\mathcal{O}_{ps} = \{\text{white}, \text{black}\}, \mathcal{O}_{ss} = \{p_{ws} \in [0,1]\},$ and $\mathcal{O}_{sp} = \{\varnothing\}$. Finally, let $S_{t,r} = w_1 p_{wa,t,r} + w_2 p_{ws,t,r}$ denote agent a's current belief state at time t and round r.

Regarding a_{sp}, once agent a states $\mathbb{E}[S_{t,r}]$, a proctor states how close agent a was to p_w. Then, agent a will enter another round with the cumulative number of personal samples n_{ps} carried from all previous rounds and the $\mu_{m,n} = \mathbb{E}[S_{m,n}]$ value from the last step in the most recent round (denoted as m and n respectively). In addition, an uncertainty penalty is applied to the variance of agent a's p_{wa} distribution. Simply put, if agent a's estimate was far from the true proportion of white balls, then it will be more uncertain about its estimate of the proportion of white balls from its own urn $p_{wa}(\alpha, \beta)$. This penalty is defined as follows: $p_{u,m,n} = (|p_w - \mathbb{E}[S_{m,n}]|)$. In essence, agent a's confidence about their estimate prior to the uncertainty penalty is reflected by $\kappa_{m,n} = \alpha_{m,n} + \beta_{m,n}$. To generate more uncertainty, we divide $\kappa_{0,n+1} = \frac{\kappa_{m,n}}{p_{u,m,n}}$. Then, $\alpha_{0,n+1} = \mu_{m,n}\kappa_{0,n+1}$ and $\beta_{0,n+1} = (1 - \mu_{m,n})\kappa_{0,n+1}$. This process allows us to increase the variance of the Beta distribution $p_{wa}(\alpha, \beta)$ without changing its expected value.

Altogether, for each time t within a round r, agent a chooses an action $a_{t,r}$, given a belief state $S_{t,r}$: essentially, agent a asks itself "Given $S_{t,r}$ and action $a_{t,r}$, what is the distribution over all possible outcomes $o_{a,t,r} \in \mathcal{O}_a$?" Afterwards, agent a uses this distribution to compute the expected utility for each action. Then, agent a chooses the action with the highest expected utility. After performing said action, agent a actually

observes $o_{t,r}$. Finally, agent a updates its belief state $S_{t+1,r}(o_{t,r}) = f(S_{t,r}, a_{t,r}, o_{t,r})$ for the next time step. The end of a round r occurs when agent a chooses a_{sp}.

In terms of utility, agent a cares about reducing the uncertainty of its estimate S, providing an accurate estimate, and minimizing cognitive cost from personal sampling. Therefore, we are left with the following functional form for utility:

$$U(S,t,r) = -\theta\sqrt{\mathrm{Var}[S_{t,r}]} - \gamma(|p_w - \mathbb{E}[S_{t,r}]|) - \tau(n_{ps,t,r}) \tag{1}$$

where $\theta, \gamma, \tau \in \mathbb{R}^+$ are the parameters for uncertainty, care for accuracy, and cognitive cost, respectively, and $n_{ps,t,r}$ corresponds to the cumulative number of personal samples up to time t in round r. From the utility function, agent a is faced with an uncertainty and cognitive sampling cost trade-off: for each sample agent a draws, they are reducing their uncertainty, but are increasing their cognitive cost. Prior research informs this functional form: Individuals want to reduce cognitive burden and uncertainty while maximizing accuracy (Hütter et al., 2016; Vasconcelos et al., 2022). Importantly, $\gamma = 0$ until the end of a round: this behaviour is due to the “proctor" not informing agent a about how close they were to the true estimate p_w until the end of the round.

Agent a optimally chooses an action at time t and round r as follows:

$$a^*_{t,r} = \begin{cases} a_{sp,t,r}, & \text{if } U(S_{t,r}) > \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\, S_{t,r}] \\ & \quad \wedge\ U(S_{t,r}) > \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\, S_{t,r}] \\ a_{ps,t,r}, & \text{if } U(S_{t,r}) \leq \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\, S_{t,r}] \\ & \quad \wedge\ \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r},\, S_{t,r}] > \mathbb{E}[U(S_{t+1,r}) \mid a_{t,r} = a_{ss,t,r},\, S_{t,r}] \\ a_{ss,t,r}, & \text{if } U(S_{t,r}) \leq \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\, S_{t,r}] \\ & \quad \wedge\ \mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r},\, S_{t,r}] \geq \mathbb{E}[U(S_{t+1,r}) \mid a_{t,r} = a_{ps,t,r},\, S_{t,r}] \end{cases}$$

Simply put, agent a chooses the action that yields the highest level of expected utility. If agent a has reached the maximum number of actions within a round or believes that social

sampling or personal sampling will decrease its current utility, then it will state its estimate for the proportion of white balls in its own urn $\mathbb{E}[S_{t,r}]$.

Agent a's expected utility with respect to social sampling is defined as follows:

$$
\begin{aligned}
\mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ss,t,r}, S_{t,r}] &= \int_0^1 U(S_{t+1,r}(o_{t,r}))\, f_{p_{ws,t,r}}(o_{t,r})\, do_{t,r} \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} U(S_{t+1,r}(o_{i,t,r}))\, f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[S_{t+1,r}(o_{i,t,r})]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[S_{t+1,r}(o_{i,t,r})] \Big| - \tau(n_{ps,t,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + w_2 p_{ws,t,r}(\alpha_{s,t+1,r}, \beta_{s,t+1,r})]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + w_2 p_{ws,t,r}(\alpha_{s,t+1,r}, \beta_{s,t+1,r})] \Big| \\
&\qquad - \tau(n_{ps,t+1,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \Bigg(-\theta\sqrt{\mathrm{Var}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r})} \\
&\overline{+ w_2 p_{ws,t,r}(\alpha_{s,t,r} + (k_{s,t,r}) o_{i,t,r}, \beta_{s,t,r} + (k_{s,t,r})\,(1 - o_{i,t,r}))]} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[w_1 p_{wa,t,r}(\alpha_{t,r}, \beta_{t,r}) + \\
&\quad w_2 p_{ws,t,r}(\alpha_{s,t,r} + (k_{s,t,r}) o_{i,t,r}, \beta_{s,t,r} + (k_{s,t,r})\,(1 - o_{i,t,r}))] \Big| \\
&\qquad - \tau(n_{ps,t,r}) \Bigg) f_{p_{ws,t,r}}(o_{i,t,r})\, \Delta o.
\end{aligned}
\tag{2}
$$

where $o_{t+1,r} \in \mathcal{O}_{ss}$, $\Delta o = \frac{1-0}{n}$, $o_{i,t+1,r} = 0 + i\,\Delta o$, and $f_{p_{ws,t,r}}(o_{t,r}) = \frac{1}{B(\alpha_{s,t,r},\beta_{s,t,r})} o_{t,r}^{\alpha_{s,t,r}-1}(1 - o_{t,r})^{\beta_{s,t,r}-1}$

In words, agent a uses its current beliefs about the proportion of white balls from social sampling, parameterized by $(\alpha_{s,t,r}, \beta_{s,t,r})$, to construct a probability density function $f_{p_{ws,t,r}}$

over all possible proportions of white balls. This distribution represents agent a's beliefs about the likelihood of observing any given proportion from agent b. For each possible proportion, agent a computes the utility associated with observing that outcome and weights it by its probability density under this distribution. Summing over all possible proportions yields the expected utility of social sampling. The way we build our utility function and our expected utility from social sampling should give rise to the second property that humans exhibit (2). More specifically, if agent a believes that the estimate provided by agent b does not reflect the proportion of white balls in agent a's urn, then it should weight $p_{ws,t,r}$ in $S_{t,r} = w_1 p_{wa,t,r} + w_2 p_{ws,t,r}$ less where $w_2 + w_1 = 1$. If $p_{ws,t,r}$ is weighed less than $p_{wa,t,r}$ in $S_{t,r}$, then agent a will gain less utility from social sampling as taking an additional social sample would not significantly reduce the uncertainty of its estimate $S_{t,r}$ in $U(S,t,r) = -\theta\sqrt{\mathrm{Var}[S_{t,r}]} - \gamma(|p_w - \mathbb{E}[S_{t,r}]|) - \tau(n_{ps,t,r})$.

Concerning agent a's expected utility with respect to personal sampling, it is

defined as follows:

$$
\begin{aligned}
\mathbb{E}[U(S_{t+1,r}(o_{t,r})) \mid a_{t,r} = a_{ps,t,r}, p_{wa,t,r}] &= \sum_{o_{t,r} \in \mathcal{O}_{ps}} U(S_{t+1,r}(o_{t,r}))\, P(o_{t,r} \mid p_{wa,t,r}, a_{ps,t,r}) \\
&= P(X_{t,r} = 1 \mid p_{wa,t,r})\, U(S_{t+1,r}(X_{t,r} = 1)) \\
&\quad + \Big(1 - P(X_{t,r} = 1 \mid p_{wa,t,r})\Big)\, U(S_{t+1,r}(X_{t,r} = 0)) \\
&= \mathbb{E}[p_{wa,t,r}]\, U\Big(w_1\, p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r}) + w_2\, p_{ws,t,r}\Big) \\
&\quad + \Big(1 - \mathbb{E}[p_{wa,t,r}]\Big)\, U\Big(w_1\, p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1) + w_2\, p_{ws,t,r}\Big) \\
&= \mathbb{E}[p_{wa,t,r}] \Bigg[- \theta \sqrt{\mathrm{Var}(p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r}))} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[p_{wa,t+1,r}(\alpha_{t,r} + 1, \beta_{t,r})] \Big| - \tau(n_{ps,t,r} + 1) \Bigg] \\
&\quad + \Big(1 - \mathbb{E}[p_{wa,t,r}]\Big) \Bigg[- \theta \sqrt{\mathrm{Var}(p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1))} \\
&\qquad - \gamma \Big| p_w - \mathbb{E}[p_{wa,t+1,r}(\alpha_{t,r}, \beta_{t,r} + 1)] \Big| - \tau(n_{ps,t,r} + 1) \Bigg].
\end{aligned}
$$

In words, agent a uses its current beliefs about the proportion of white balls from personal sampling, parameterized by $(\alpha_{t,r}, \beta_{t,r})$, to determine the probability of drawing a white or black ball, where $P(\text{white}) = \mathbb{E}[p_{wa,t,r}]$ and $P(\text{black}) = 1 - \mathbb{E}[p_{wa,t,r}]$. For each possible outcome, agent a computes the utility associated with observing that ball and weights it by its probability. Summing over both outcomes yields the expected utility of personal sampling. The way we constructed our utility function and our expected utility from personal sampling should give rise to the third property that humans exhibit (3). More specifically, when agent a faces a low cognitive cost from personal sampling, taking an additional personal sample becomes more attractive, as agent a incurs little loss in utility while helping to reduce uncertainty in the estimate.

Finally, by the construction of our model, property (5) can be exhibited. Suppose

that agent a faces a high cognitive cost and an estimate from agent b that reflects the true proportion of white balls in agent a's urn. It follows that agent a's $p_{ws,t,r}$ in $S_{t,r} = w_1 p_{wa,t,r} + w_2 p_{ws,t,r}$ will be weighed approximately the same as $p_{wa,t,r}$, such that $w_2 \approx w_1$. Crucially, given that cognitive cost is high, agent a incurs a major utility loss from personal sampling. Fortunately, given that $w_2 \approx w_1$, agent a can significantly reduce the uncertainty of its estimate $\mathbb{E}[S_{t,r}]$ by social sampling without incurring any cognitive cost. Therefore, agent a should rely more on social sampling than personal sampling. Moreover, by the construction of our model, property (6) can be exhibited. Suppose that agent a faces a low cognitive cost and an estimate from agent b that is far from the true proportion of white balls in agent a's urn. It follows that agent a's $p_{ws,t,r}$ in $S_{t,r} = w_1 p_{wa,t,r} + w_2 p_{ws,t,r}$ will be weighted much less than $p_{wa,t,r}$, such that $w_2 \approx 0$. Crucially, given that $w_2 \approx 0$ agent a cannot meaningfully reduce the uncertainty of its estimate $\mathbb{E}[S_{t,r}]$ by social sampling. Fortunately, given that cognitive cost is low, agent a can significantly reduce the uncertainty of its estimate $\mathbb{E}[S_{t,r}]$ by personal sampling without incurring much cognitive cost. Thus, agent a should rely more on personal sampling than social sampling. Both these properties (5, 6) give rise to property four (4).

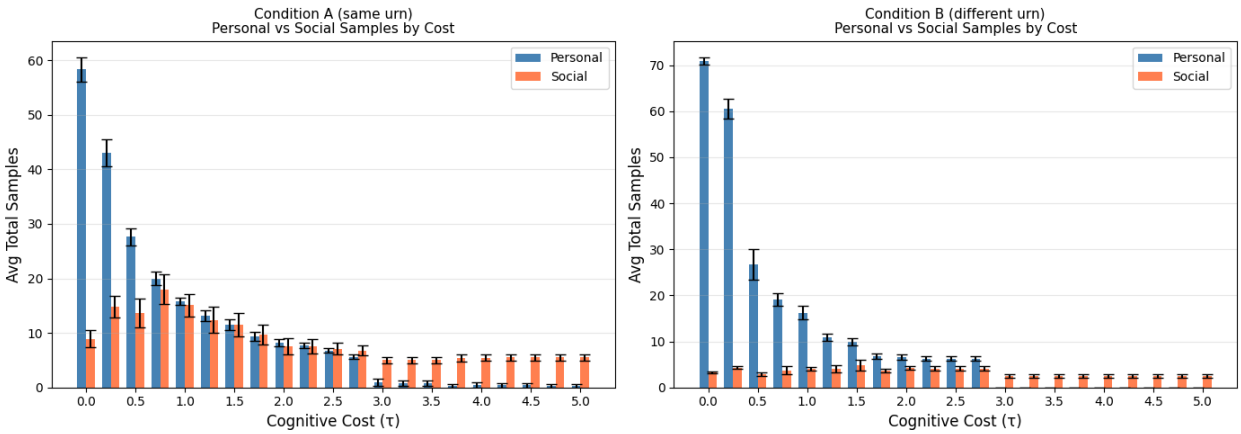
Simulation

To keep our analysis simple and make sure that human subjects can complete our experimental design in the future, agent a was presented with five rounds. Within each of these rounds, agent a could perform up to 15 actions; if agent a decided to state its estimate, then a round ended. We modulated the cognitive cost from personal sampling τ from 0 to 5 with 0.25 increments; we chose this span as agent a's total number of personal and social samples across all rounds usually stabilize beyond $\tau = 4$ (See Appendix F and G). Agent a was exposed to two conditions: in condition A, the urns of agents a and b each contained 30% white balls, and in condition B, agent a's urn contained 30% white balls, whereas agent b's urn contained 80% white balls. Therefore, when accounting for cognitive cost and urn composition manipulations, we truly had 42 conditions in total. For each

condition, we ran our model 15 times and constructed bar plots for the average total number of personal and social samples that agent a performed across all rounds. From preliminary simulations, we observed that the total number of personal and social samples was substantially less sensitive to variation in θ (uncertainty aversion) compared to τ (cognitive cost). A similar pattern was observed for γ (accuracy weighting). For the main results table, we fixed $\theta = 70$ and $\gamma = 100$. To assess the robustness of our findings, we systematically varied each parameter while holding the other constant. Specifically, we varied θ from 0 to 120 in increments of 10 while fixing $\gamma = 100$ (see Appendix F), and varied γ from 0 to 120 in increments of 10 while fixing $\theta = 70$ (see Appendix G). Across both analyses, the qualitative patterns of results remained unchanged. Finally, to examine behavior in more extreme regimes, we also evaluated larger parameter values ($\theta, \gamma \in 200, 500$), and again found no substantive deviations from the main patterns (See Appendices F and G). For agent a to not enter another round, $|p_w - \mathbb{E}[S_{t,r}]| < 0.0001$ as we wanted to make sure that our agent had the correct answer. Finally, agent a's initial beliefs about its personal urn and agent b's estimate were $\alpha = 2, \beta = 2$ and $\alpha_s = 2, \beta_s = 2$, respectively ($Beta(2, 2)$) for both personal and social samplings).

Results

Again, we obtained our results by fixing θ to 70 and γ to 100. The left graph concerns the results from condition A, whereas the right graph describes the results from condition B. For each graph, we modulated τ from 0 to 5 with 0.25 increments. Thus, agent a was exposed to 21 conditions within each graph. Finally, for each condition, we computed the average total number of personal and social samples that agent a performed across 15 different seeds.



Discussion

Across both graphs, the average total number of personal samples decreases as the cognitive cost of personal sampling increases. This pattern is consistent with the third behavioural property exhibited by humans (3). Moreover, the average total number of social samples decreases when the urns are different compared to when the urns are the same, which aligns with the second behavioural property (2). Regarding the graph on the left, we see that agent a prefers to perform more social samples than personal samples when agent b's estimate can be relied upon and the cognitive cost is high. Importantly, this pattern reflects the fifth behavioural property (5). Concerning the graph on the right, we see that agent a prefers to draw personal samples more than social samples when agent b's estimate cannot be relied upon and the cognitive cost is low. This pattern is congruent with the sixth behavioural property (6). Given that the fifth and sixth behavioural properties are exhibited, we can conclude that agent a's behaviour satisfies the fourth behavioural property (4). Finally, by construction of the w_2 and w_1 weights, the model satisfies the first behavioural property (1). Crucially, these behavioural patterns do not change when we vary uncertainty aversion and need for accuracy (See Appendix F and G). Therefore, we can conclude that our model satisfies all specified properties.

Concerning limitations, the model provides a low resolution account of human behavior. Even though it reproduces human behavior in terms of general patterns

(Armitrage and Redsaw, 2025), we have not yet tested whether it can reliably predict choices from real human data. In addition, we treat cognitive cost as a constant, but in reality, it should vary as the agent moves through rounds. For instance, an agent who has done a lot of personal sampling will get more fatigue of doing more, so the cost will increase (Jeong et al., 2022). Finally, our utility function only includes uncertainty, accuracy, and cognitive cost, but we have not incorporate things like emotions, motivation, and other human features that may affect decision-making.

Regarding future directions, the next step is to run the experiment on human participants and estimate our model's parameters from actual choices. We would need to change cognitive cost such that it varies over time across rounds. For example, we can increase the cost proportional to the number of personal samples or set it such that the cost may decrease in later rounds as the agent gets better with practice. Another useful extension would be to incorporate emotion and first-impression effects. We could model emotion as a factor that shifts existing parameters. For instance, anxiety might increase the cognitive cost parameter or decrease uncertainty aversion, while overconfidence might reduce the need for accuracy. First impression effects could be modeled by giving early social samples more weight, so that it can influence the agent's belief more. Future experiments could impose a measurable cognitive cost by requiring participants to solve a short arithmetic question before each personal sample. This would make cognitive cost observable and allow us to examine whether higher effort directly shifts behavior toward social sampling.

All in all, our Bayesian model that can make decisions can exhibit behavioural properties: $1, 2, 3, 4, 5,$ and 6, meaning that it can approximate the trade-off between cognitive cost from solving problems on your own and accuracy from asking other people for the solution.

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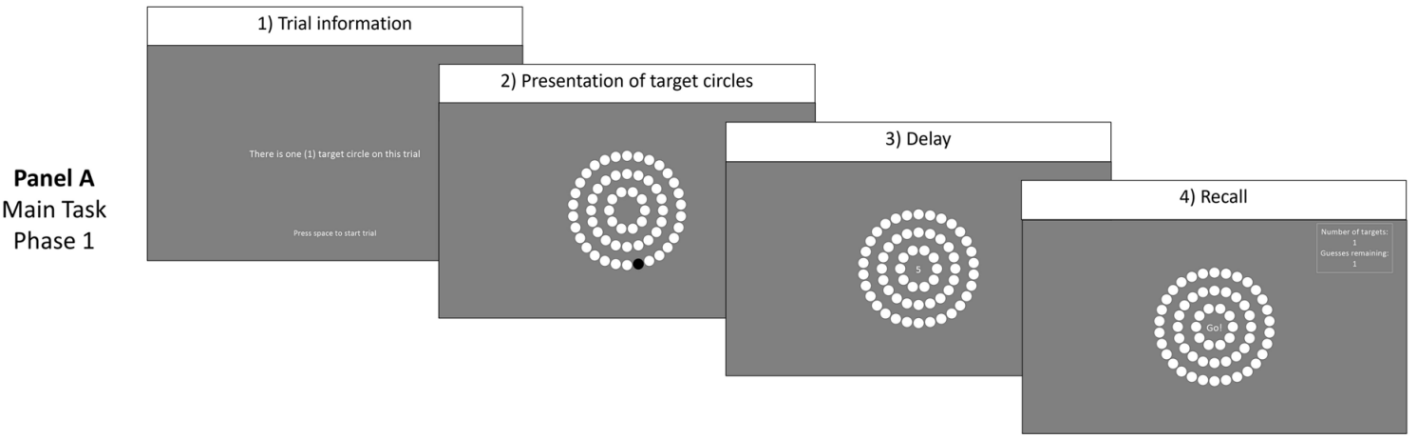
Appendix A

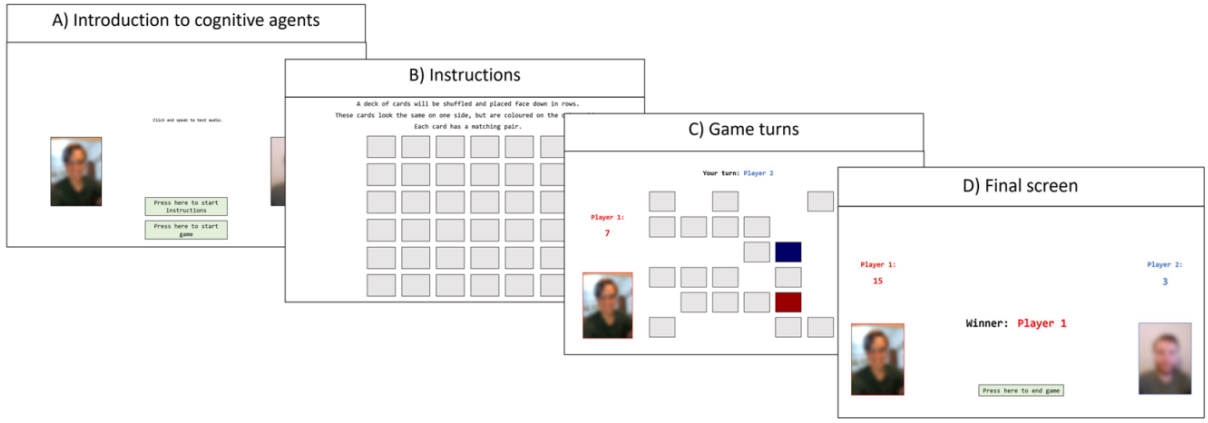
First Phase



Appendix B

Introduction to the Competent Agent and the Incompetent Agent

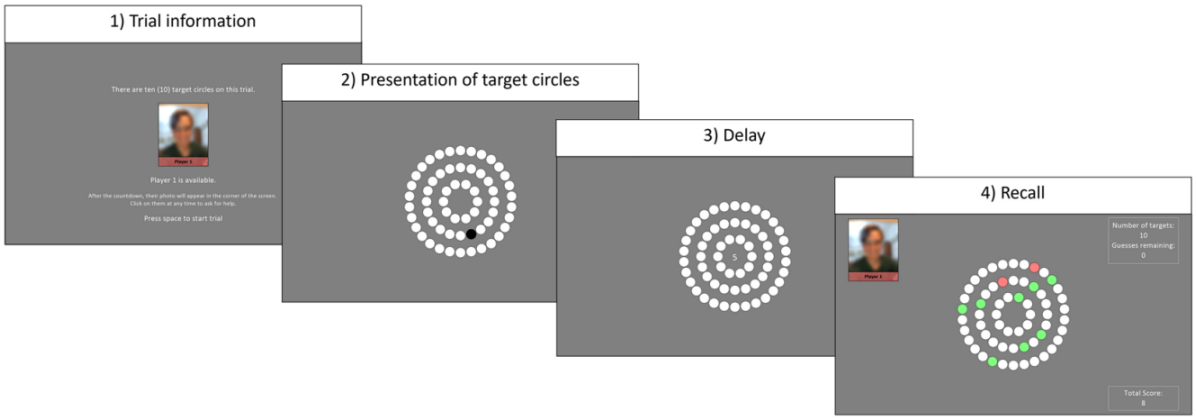
Panel B
Memory
Competence of
Helpers



Appendix C

Second Phase

Panel C
Main Task
Phase 2



Appendix D

Teammate Contributions

- Andrew: Created the model, wrote and reviewed the code, edited the paper, heavily contributed to the literature review.
- Theo: Debugged the code and implemented revision, implemented the weight distribution of personal and social sample, edited the paper, provided responses to classmate questions.
- Ip: Helped run simulations, analyzed the simulation results, prepare the presentation and response to questions, revised and edited the paper.

Appendix E

Questions from Classmates

- Question 1: How is “difficulty" adequately determined? On what terms?
  - Answer: In our model, we think of “difficulty" as how hard it is for the agent to reach a confident and accurate estimate of the urn’s true proportion. In general, if the Bayesian agent performs less samples in total, then it is more difficult for it to estimate the true proportion of white balls in its urn.
- Question 2: Our emotions often heavily influence our decision-making ability. So how does/can your model account for this?
  - Answer: Currently, our model does not explicitly include emotions, so we count that as a limitation. For now, the agent makes decisions only by maximizing expected utility based on uncertainty, expected error, and cognitive cost. Also, it would interesting to look at how different types of emotions, such as anger and sadness, affect decision making.
- Question 3: How would you expect the behavior to a problem with moderate difficulty?
  - Answer: Assuming that difficulty refers to the cognitive cost from personal sampling, we would expect less personal sampling if we increased the cognitive cost.
- Question 4: What is the prior to your model?
  - Answer: There are two priors in our model. One for personal sampling and one for social sampling. They are both Beta distributions with $\alpha = 2$ and $\beta = 2$.

- Question 5: How would this human behavior be empirically measured?
  - Answer: We can measure this by observing how often participants choose personal sampling versus social sampling across many rounds. We would see how the choice changes as the task difficulty varies and how they rely on competent versus incompetent social sources.
- Question 6: Would it be possible to consider varying cognitive cost of doing different tasks for different people?
  - Answer: Yes. We currently use a fixed cognitive cost for simplicity, so that we can clearly identify the trade-off between cognitive cost and social agent accuracy. But, in reality, cognitive cost would be different for different tasks and people. For instance, an individual with higher mental math skills may have a lower cognitive cost for the same task.
- Question 7: Is there a threshold in cognitive cost where the decision strategy changes significantly, rather than gradually?
  - Answer: This threshold does exist; however, it depends on the combination of uncertainty aversion (θ), cognitive cost (τ), and need for accuracy (γ) parameters. We did not explore this idea; however it would be interesting to investigate in future research.
- Question 8: How is the cognitive cost determined and why we can fix a constant?
  - Answer: In our model, cognitive cost is not directly measured from human data. Instead, it is a theoretical parameter that represents the mental effort of doing personal sampling. We assign it as a fixed constant so that we can clearly isolate its effect on behavior and study the trade-off between cognitive cost and social agent accuracy.

- Question 9: Sometimes people are getting wrong results from personal sample, and right result from social's, but the actual situation is social has a much higher probability of wrong sample, but people can get behavioural reliance based on first impression.
  - Answer: We are not sure what your question is.
- Question 10: Would you keep your model or opt for an alternative one to estimate cognitive cost from real human behavior?
  - Answer: For our current goal, we would keep the model as is, as it focuses on capturing the trade-off we care about in a clean and interpretable way. But if the goal were to estimate cognitive cost from real human behavior more precisely, we would probably extend it more by including time-dependent cognitive cost.
- Question 11: How did you decide on your prior?
  - Answer: The priors for personal sampling and social sampling came from another research project that Andrew is currently working on. He asked young adults to express their prior beliefs about the proportion of white balls in an urn via a discrete probability distribution. He also asked them to express their prior beliefs about the proportion of white balls stated by a social agent via a discrete probability distribution. He then fit a Beta distribution to these probability distributions and obtained $\alpha = 2$ and $\beta = 2$ for personal and social sampling.
- Question 12: How is cognitive cost measured?
  - Answer: In our model, cognitive cost is not directly measured from data. It is represented as a parameter in the utility function that penalizes personal sampling, so each additional personal sample depletes utility.

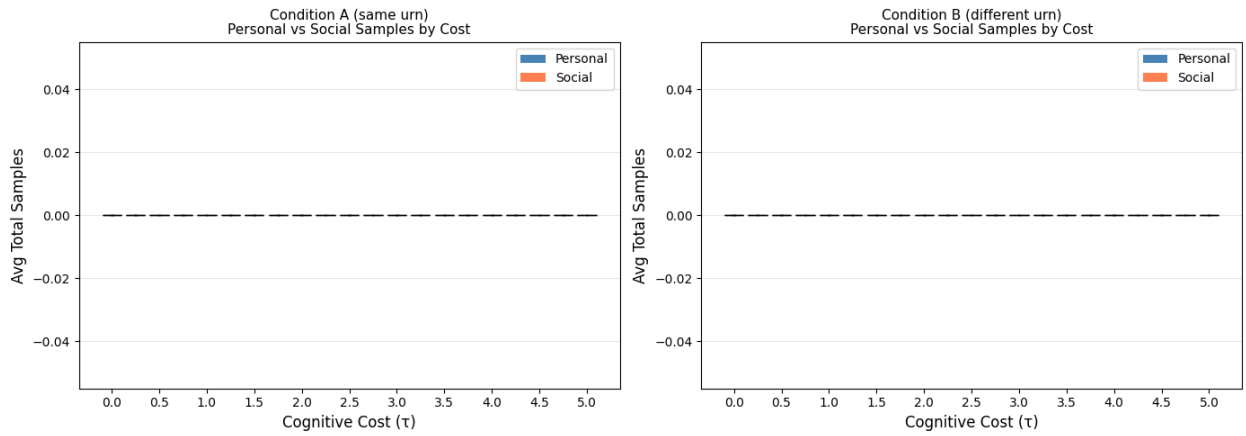
- Question 13: Are there any other probability models that can be used to model cognitive cost?
  - Answer: Yes. Other probability models could also be used to represent cognitive cost. For example, a Bernoulli or Binomial model could treat cognitive cost as increasing the probability of an error or premature stopping, while a Gaussian model could represent cognitive cost as added noise in the agent's internal computations. We chose a utility penalty because it is simple, interpretable, and fits naturally within an expected-utility framework, but alternative probabilistic formulations would be a useful direction for future work.
- Question 14: Based on the extent of successful replication of the reference psychological experiment setup by your model, how well do you believe your model can recapitulate more abstract scenarios of this cognitive phenomena?
  - Answer: We are not sure what you are asking us.
- Question 15: How does setting a fixed cognitive cost impair the predictive ability of your model?
  - Answer: Our research did not look into that question; however, this would be a good future direction.

Appendix F

Simulations Where We Modulated θ (Uncertainty Aversion) and Fixed $\gamma = 100$ (Need for Accuracy)

Figure F1

θ = 0



For this figure, please note that setting $\theta = 0$ essentially means that agent a is uninterested in reducing the uncertainty of its estimate so it will state its proportion of white balls for each round within trying to collect any new data.

Figure F2

θ = 10

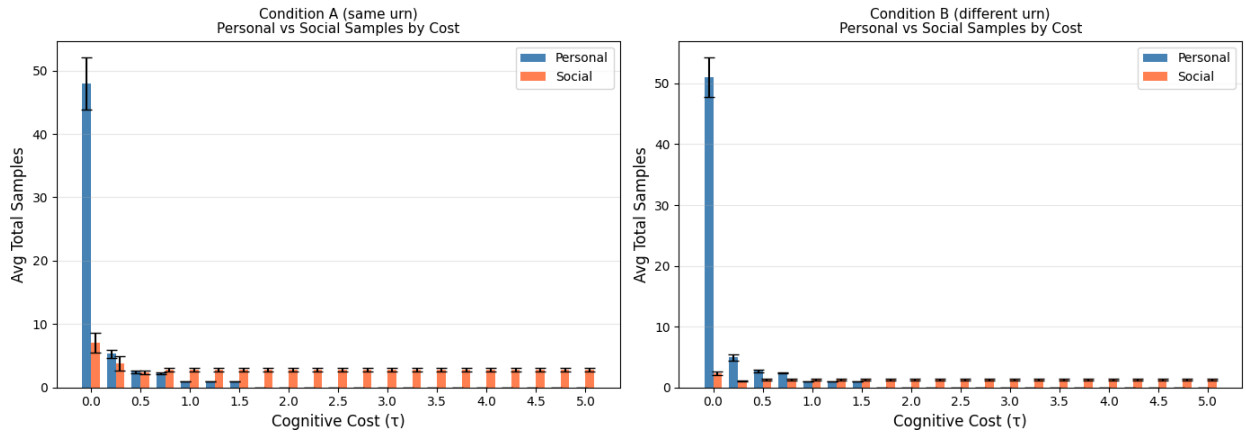


Figure F3

θ = 20

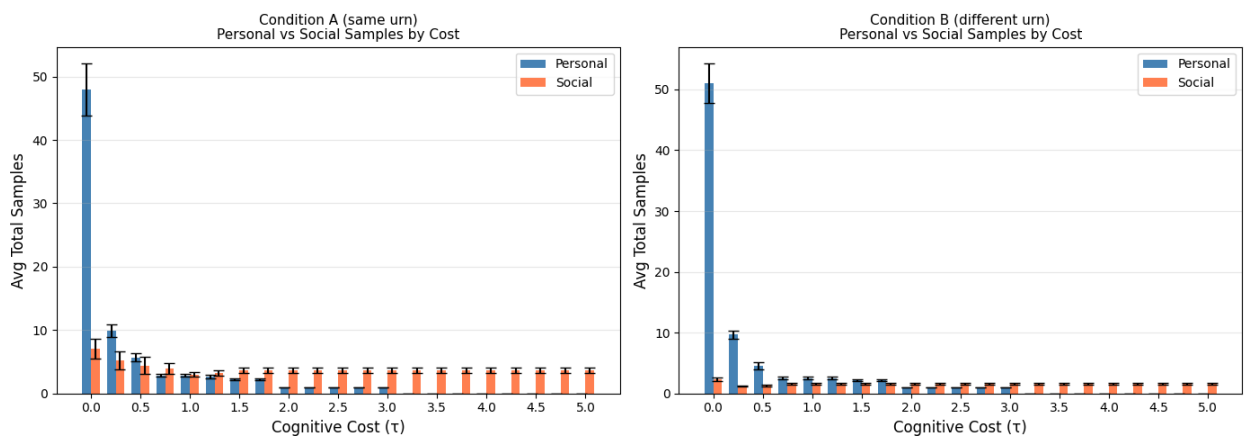


Figure F4

$\theta = 30$

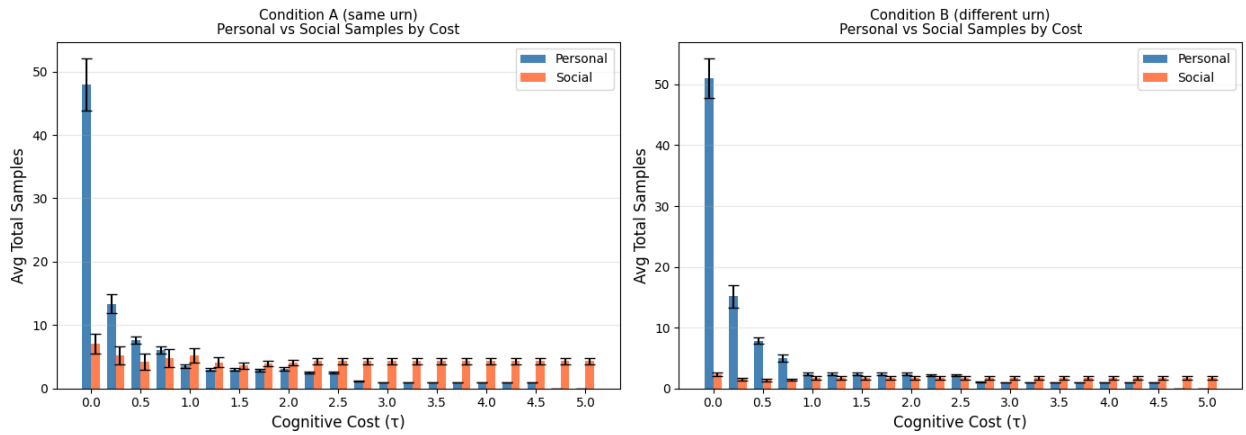


Figure F5

$\theta = 40$

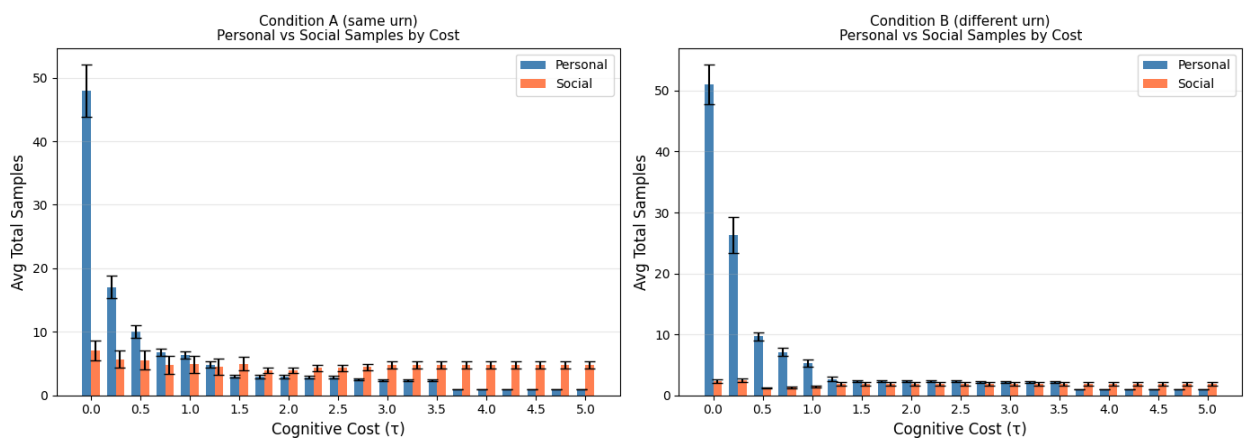


Figure F6

$\theta = 50$

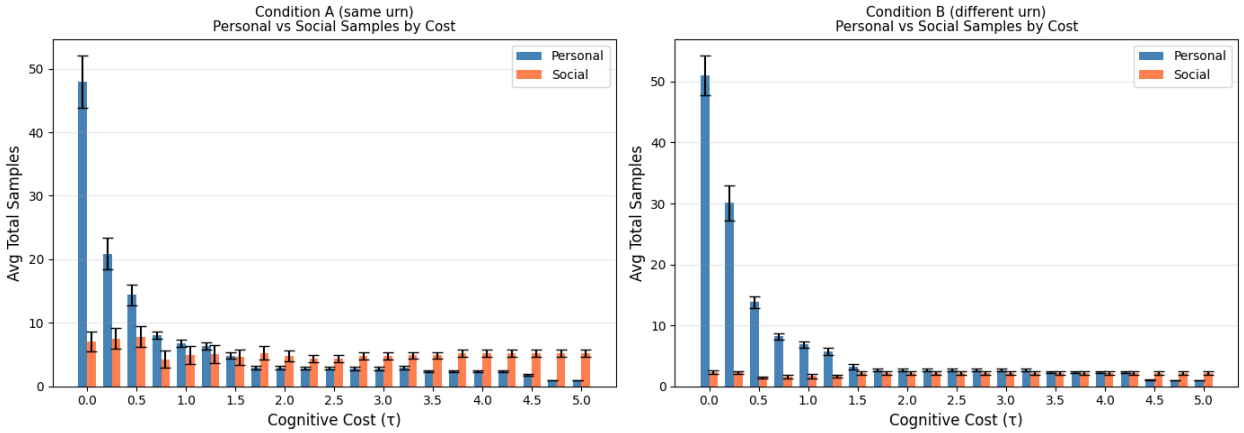


Figure F7

$\theta = 60$

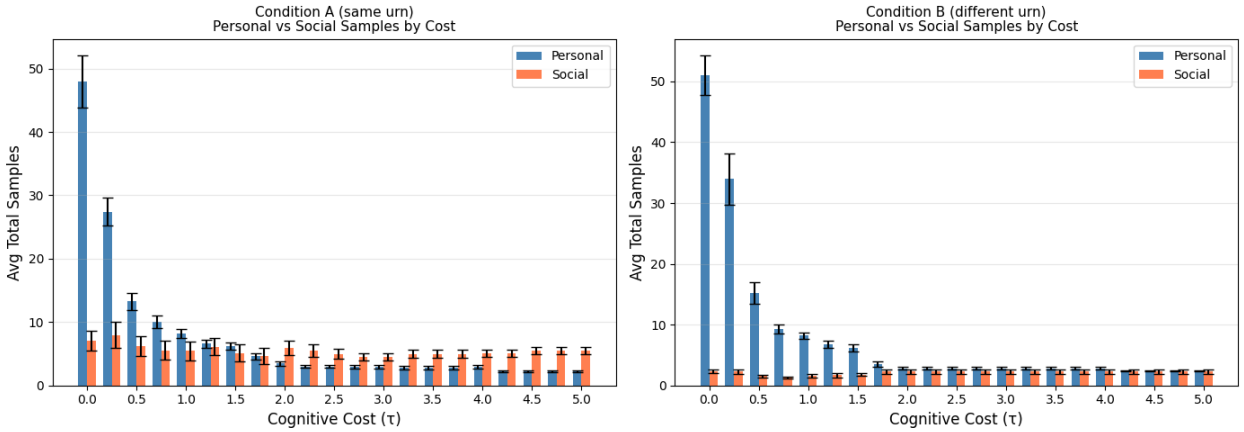


Figure F8

$\theta = 70$

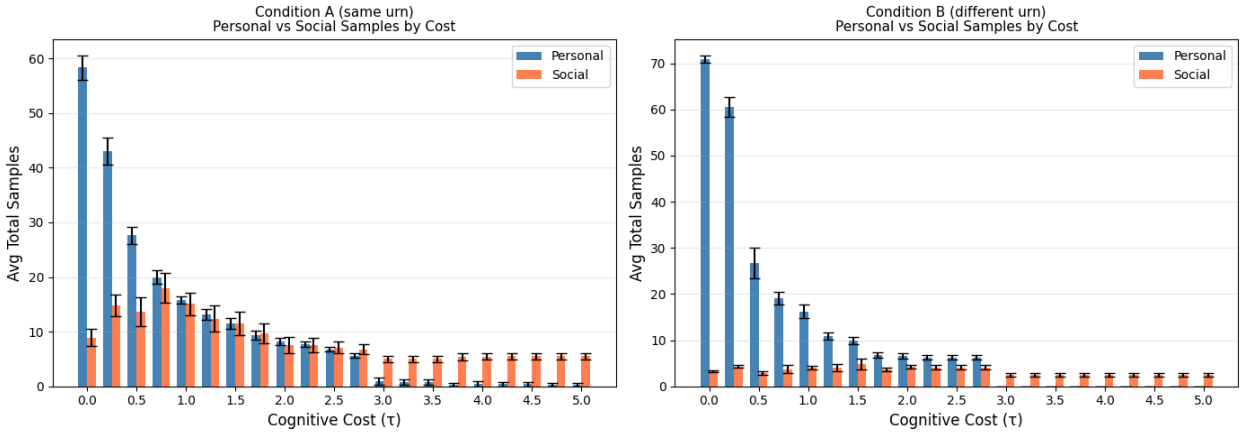


Figure F9

$\theta = 80$

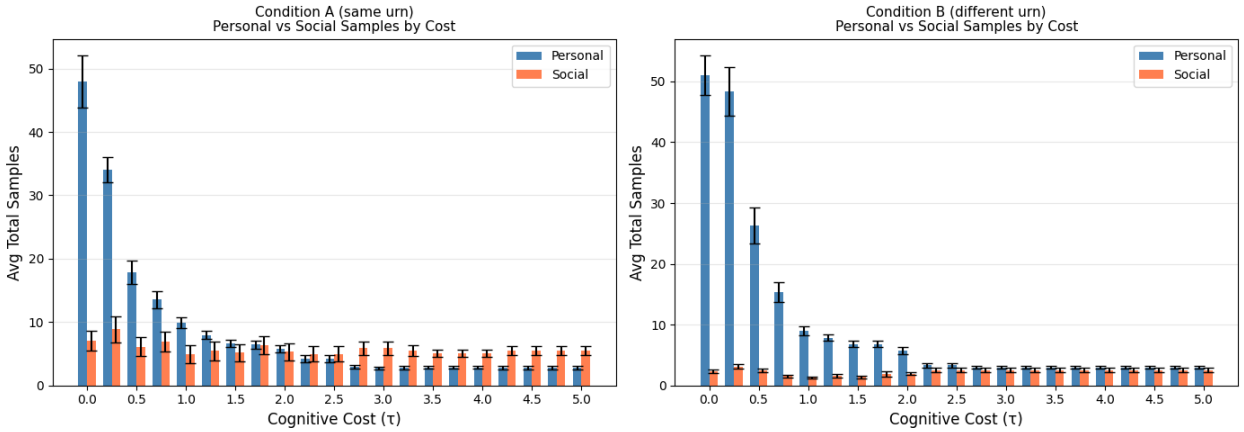


Figure F10

θ = 90

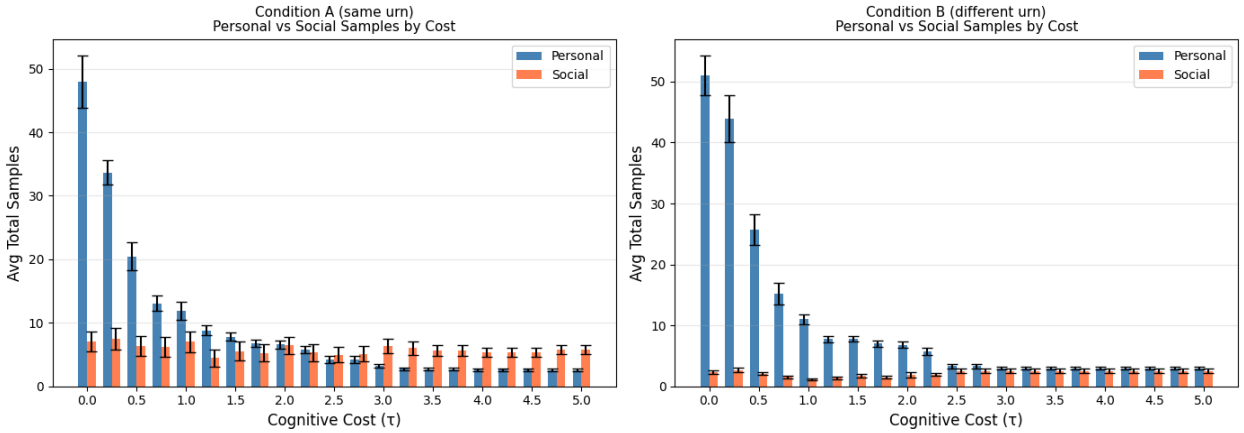


Figure F11

θ = 100

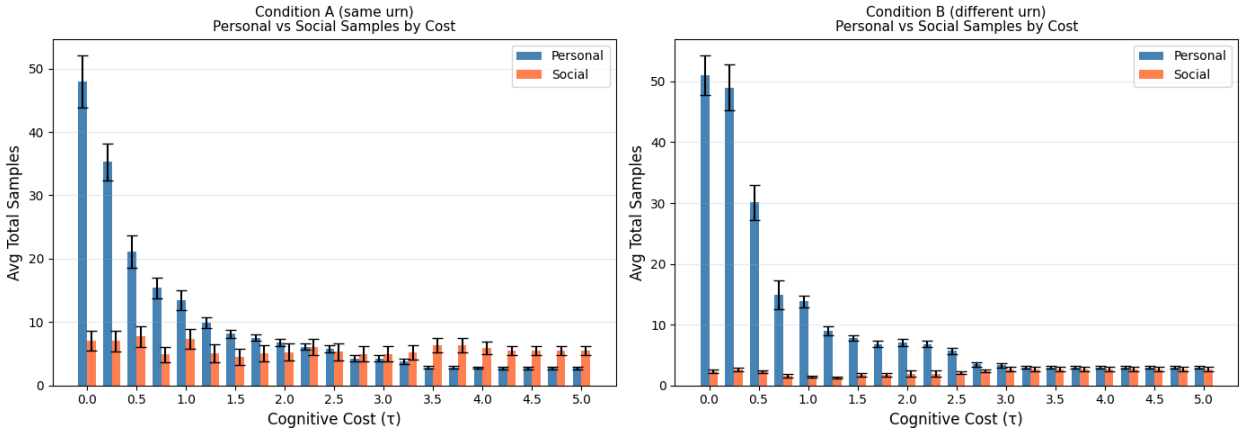


Figure F12

θ = 110

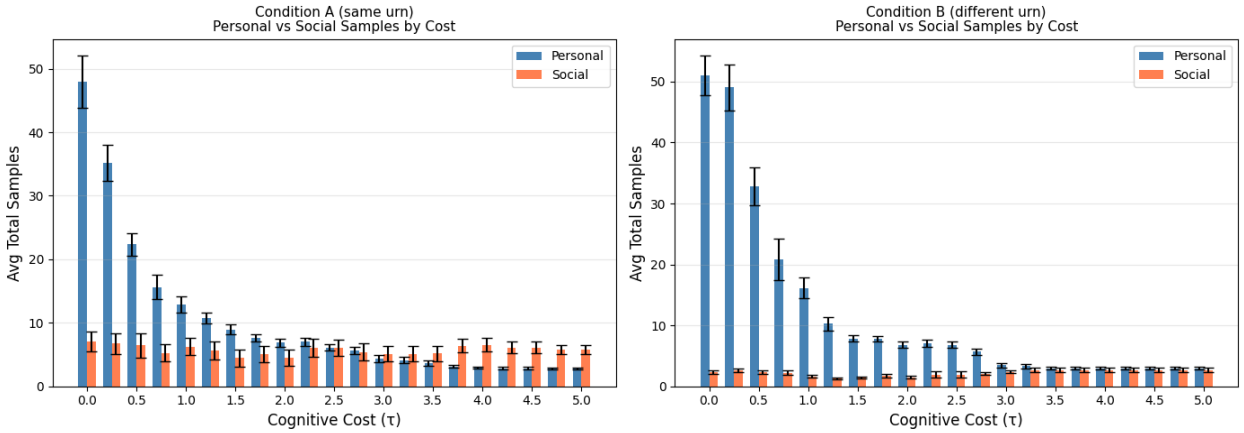


Figure F13

θ = 120

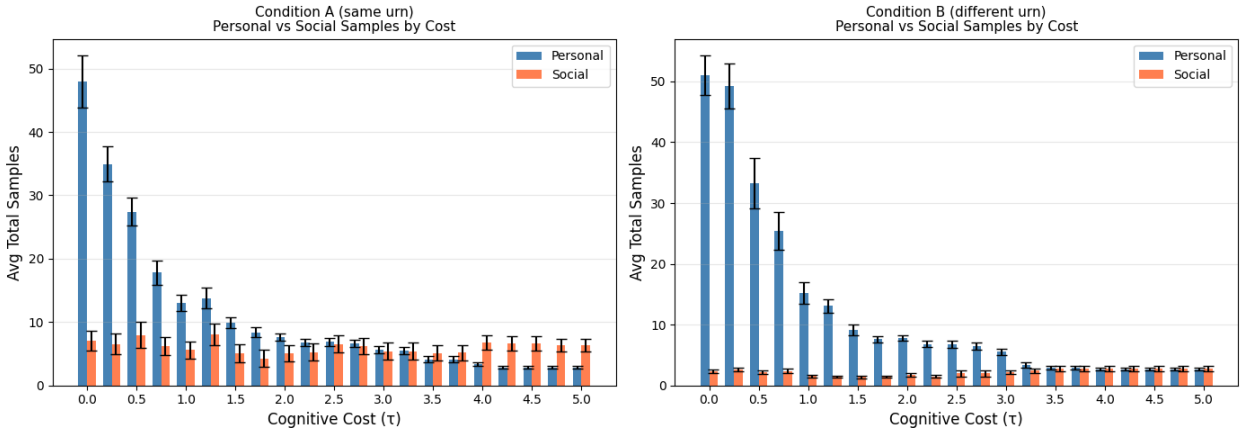


Figure F14

θ = 200

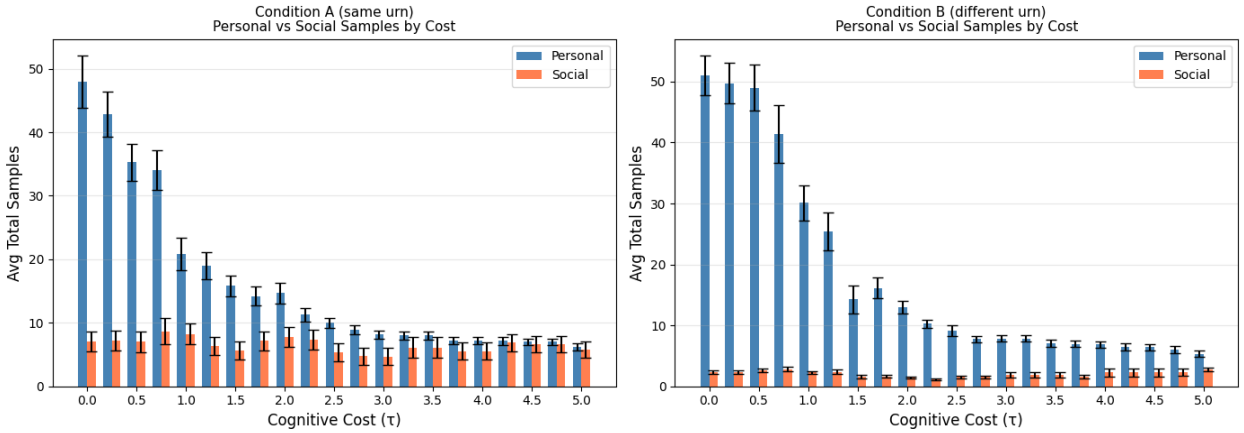
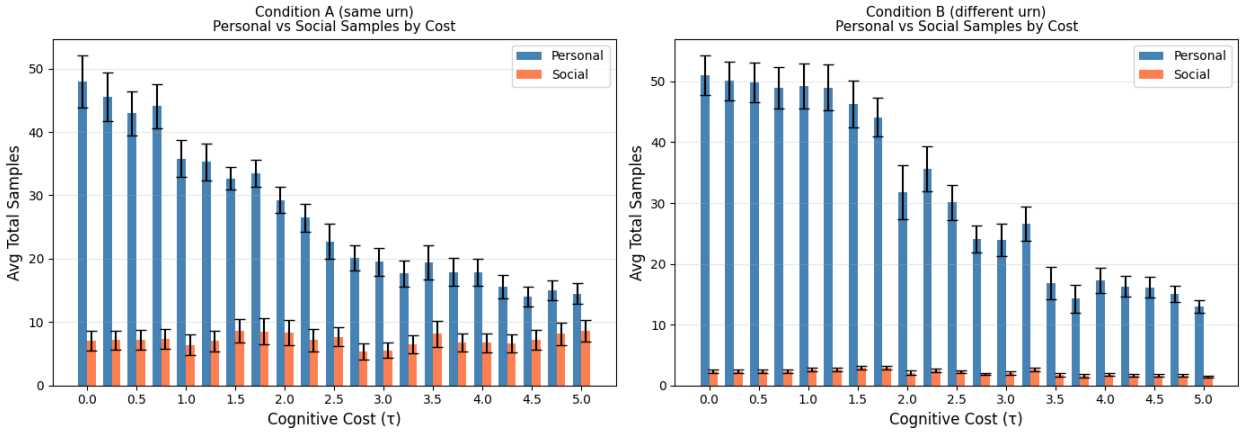


Figure F15

θ = 500



Appendix G

Simulations Where We Modulated γ (Need for Accuracy) and Fixed $\theta = 70$ (Uncertainty Aversion)

Figure G1

γ = 0

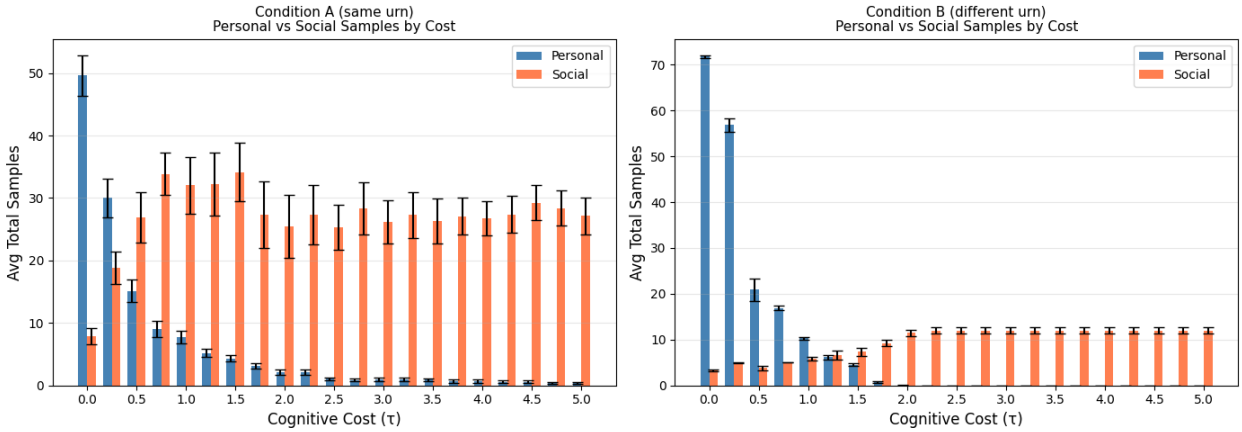


Figure G2

γ = 10

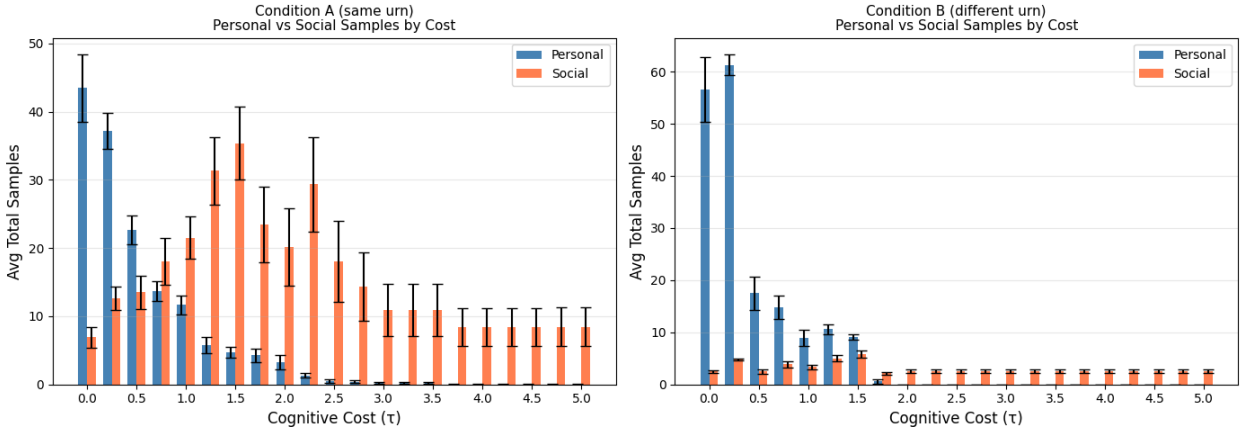


Figure G3

$\gamma = 20$

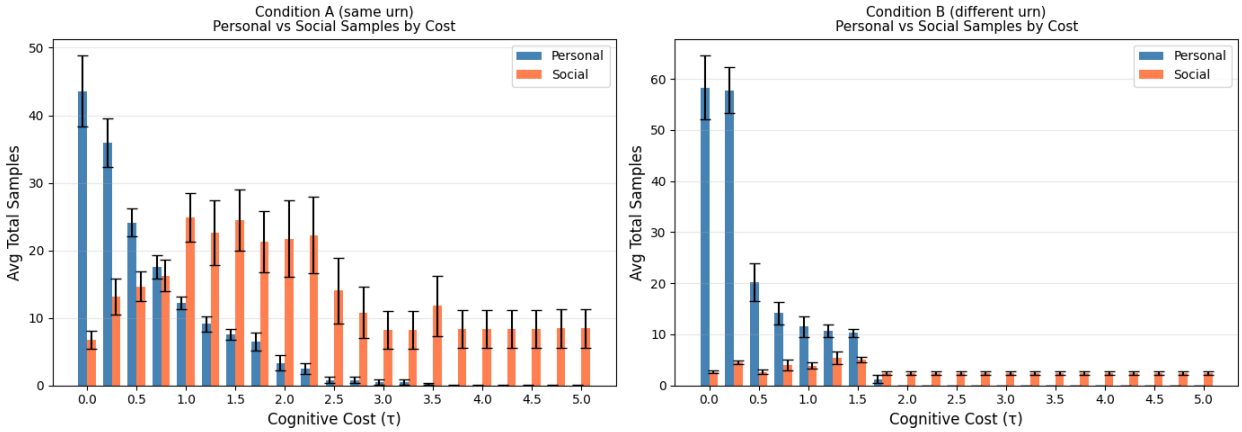


Figure G4

$\gamma = 30$

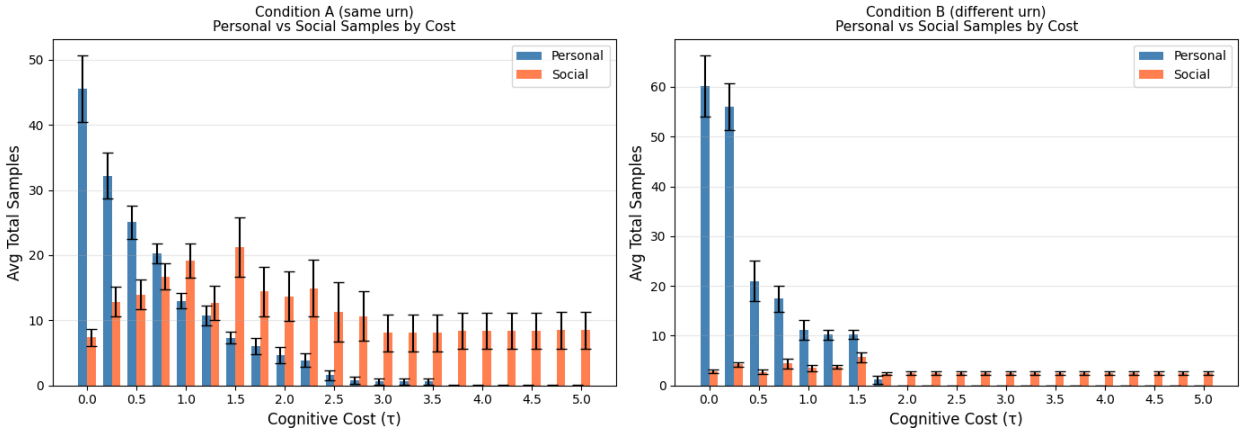


Figure G5

$\gamma = 40$

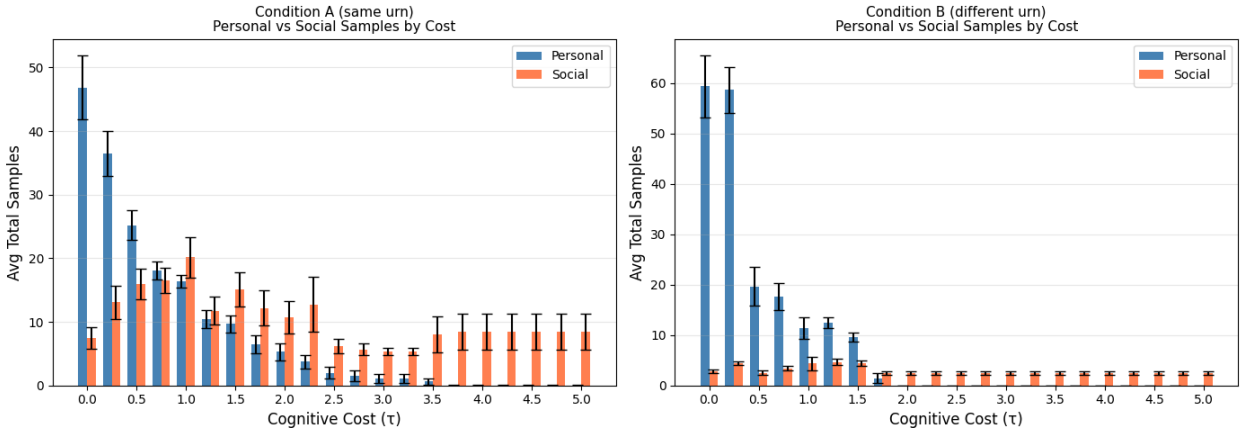


Figure G6

$\gamma = 50$

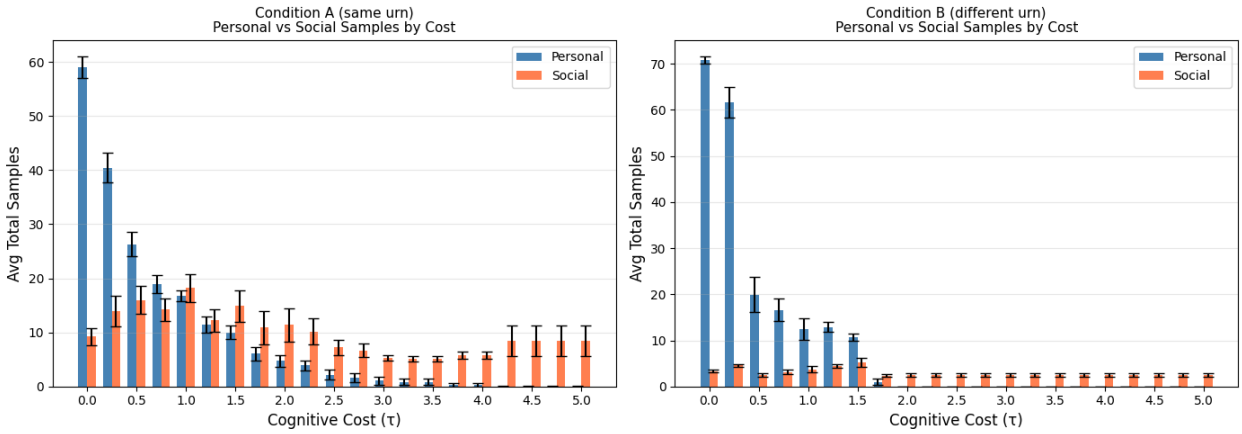


Figure G7

$\gamma = 60$

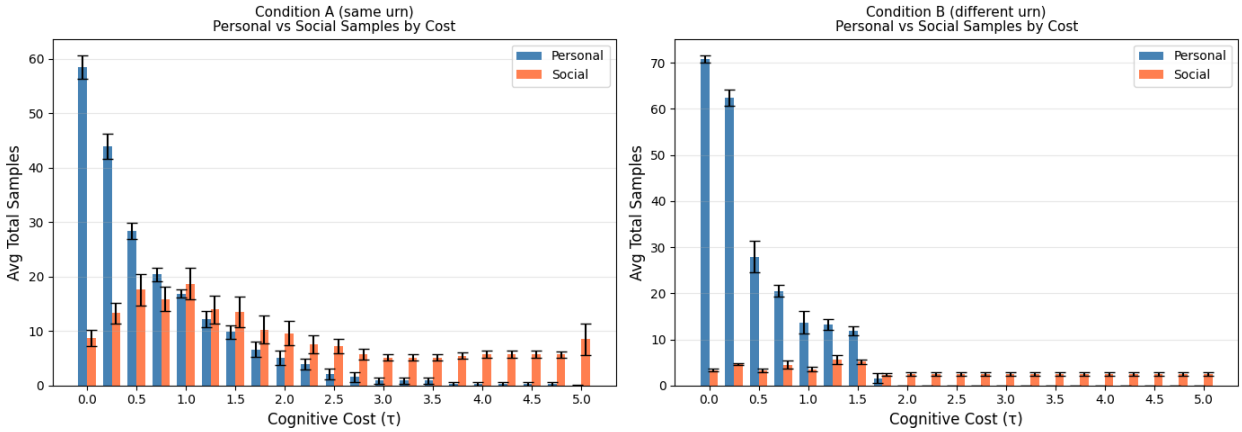


Figure G8

$\gamma = 70$

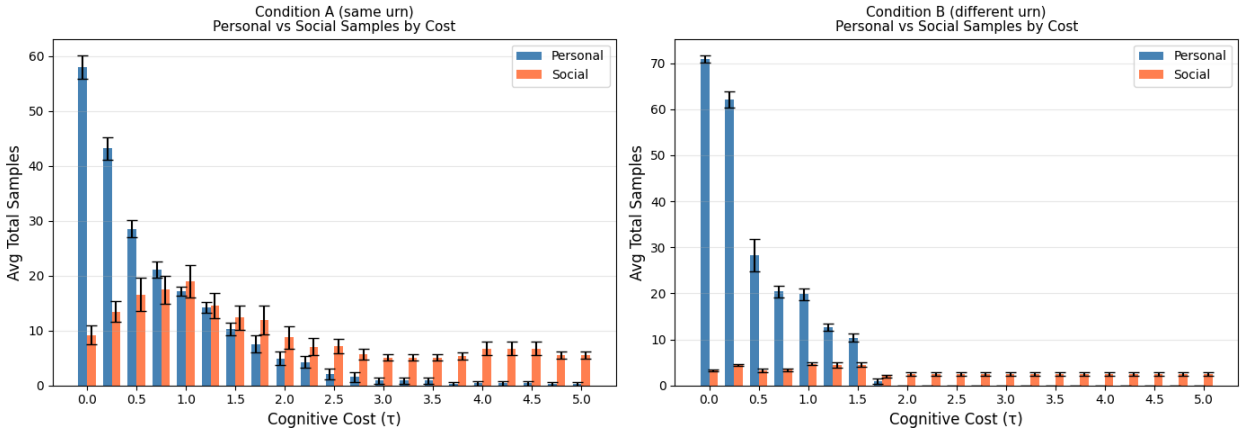


Figure G9

$\gamma = 80$

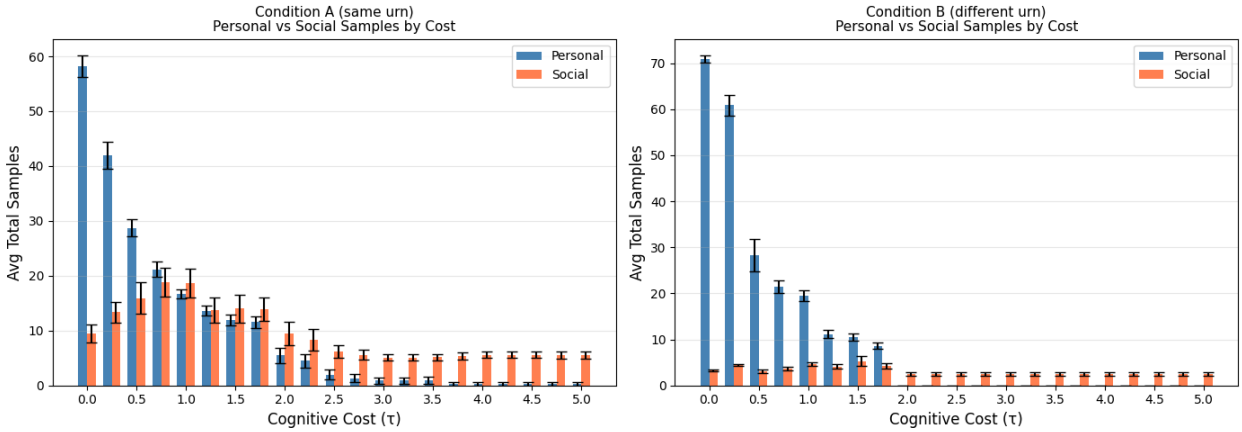


Figure G10

$\gamma = 90$

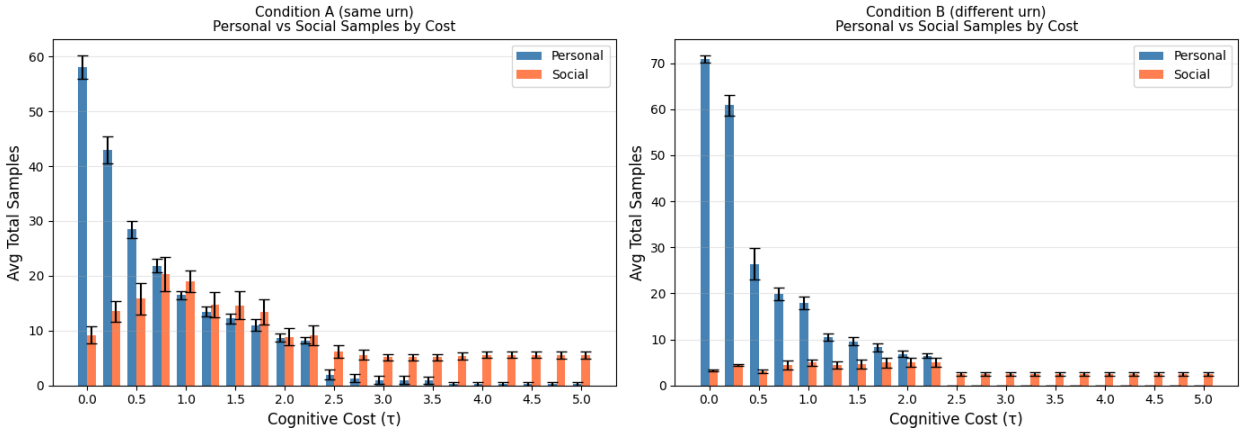


Figure G11

γ = 100

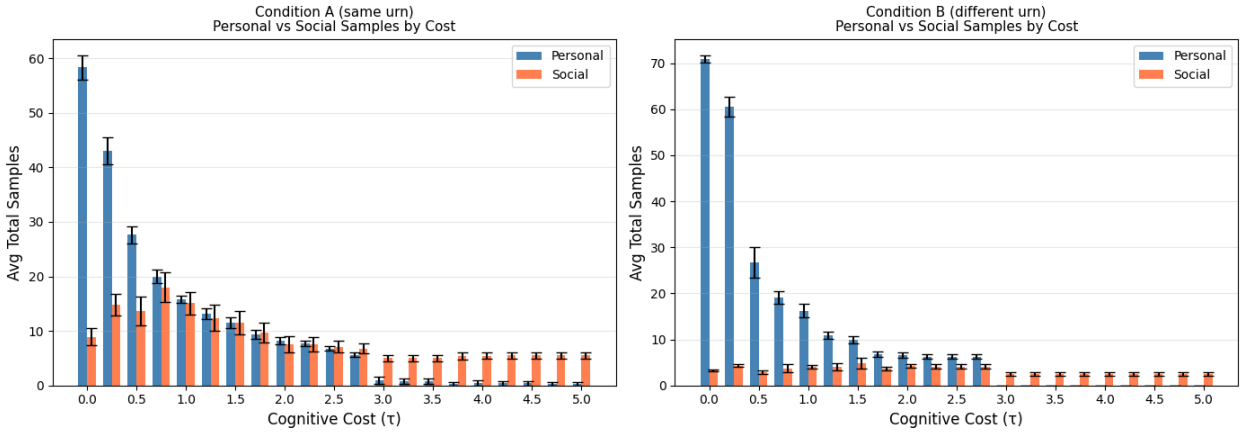


Figure G12

γ = 110

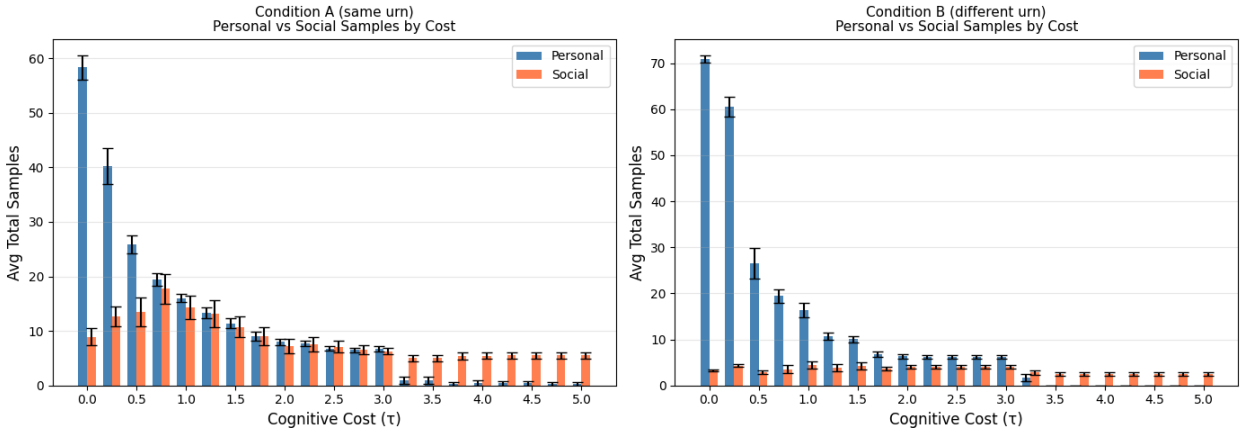


Figure G13

$\gamma = 120$

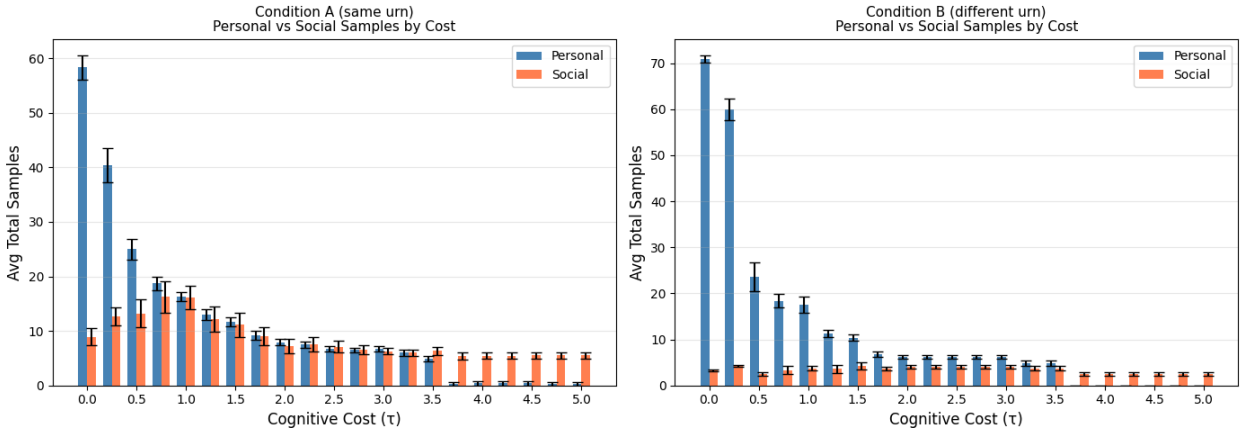


Figure G14

$\gamma = 200$

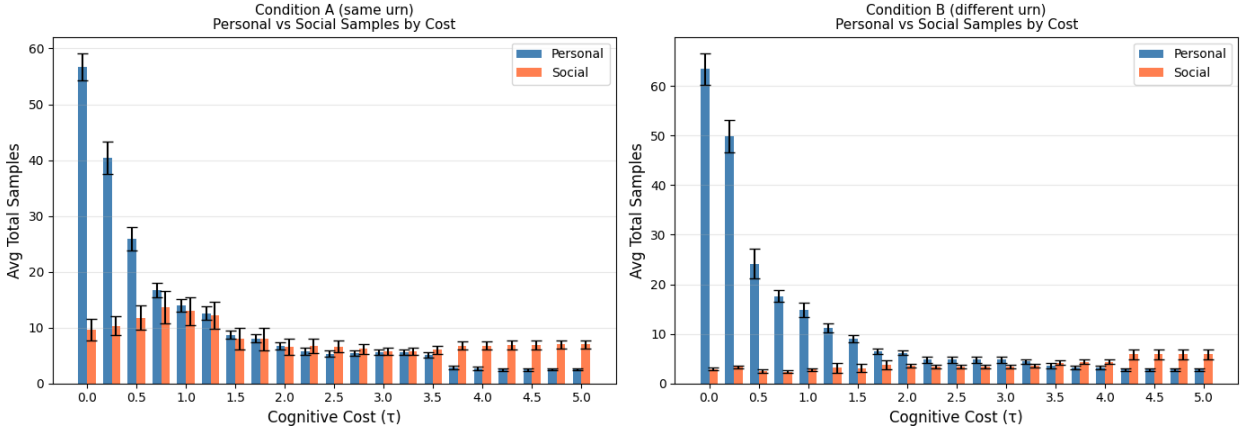
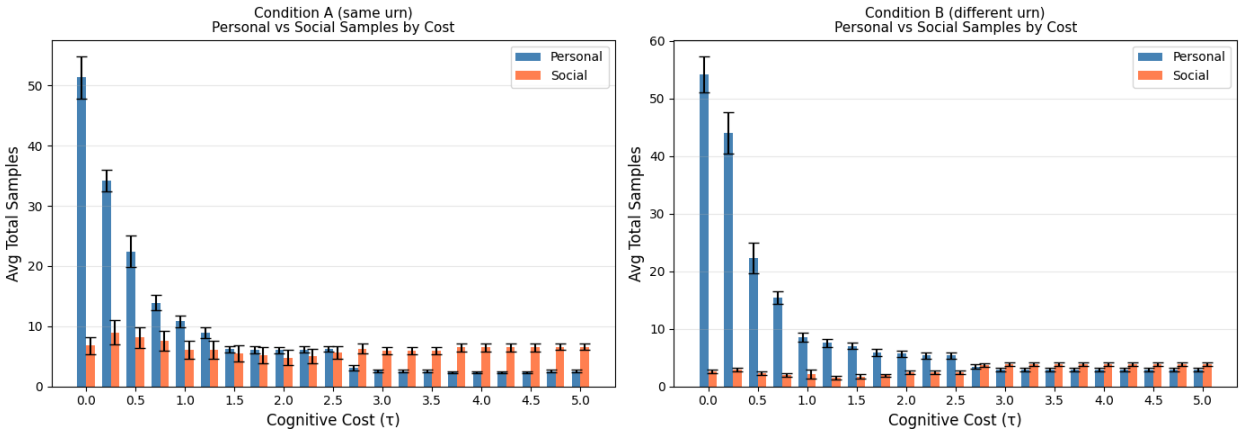


Figure G15

$\gamma = 500$